Phototrophs: These are organisms that utilize light (radiant energy) as their energy source.

Chemotrophs: These are organisms that obtain energy by oxidation of organic or inorganic compounds.

Microorganisms are classified into **Lithotrophs** and **Organotrophs** based on the source from which they extract electrons. Lithotrophs are organisms that use reduced inorganic substances as their electron source whereas Organotrophs obtain electrons from organic compounds (Table 6.1).

All microorganisms fall into any one of the four nutritional classes based on their primary source of carbon, energy and electrons.

1. **Photoautotrophs:** Eukaryotic algae, Cyanobacteria (Blue Green Algae) (Figure 6.1) and Purple and Green Sulphur bacteria belong to this class. They are capable of using light energy and have carbondioxide as the sole source of carbon.

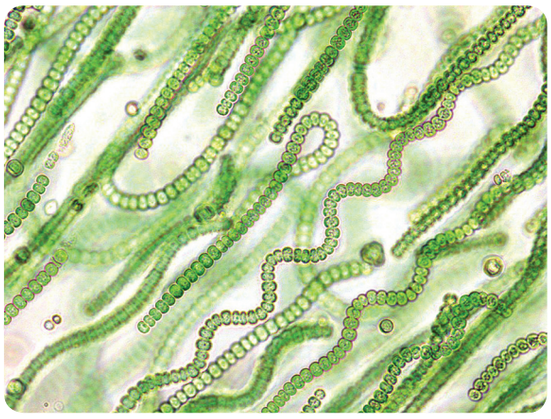


Figure 6.1: Microscopic view of *Cyanobacteria*

2. **Photoheterotrophs:** These organisms make use of light as energy source and organic compounds as electron and carbon source. Example: Purple and Green Non sulphur bacteria
3. **Chemoautotrophs:** These are ecologically important microorganisms. They oxidize inorganic compounds like nitrate, iron and sulphur to obtain energy and electrons.
4. **Chemoheterotrophs:** These organisms use organic compounds to satisfy their needs of energy, electron and carbon. (Table 6.2)

Table 6.1: Classification of microorganism based on carbon, energy and electron sources

Carbon, Energy and Electron sources	
Carbon sources	
Autotrophs	CO_2 as sole carbon source
Heterotrophs	Organic substances from other organisms
Energy sources	
Phototroph	Light energy
Chemotrophs	Chemical energy source (Organic or Inorganic)
Electron sources	
Lithotrophs	Reduced inorganic substances
Organotrophs	Organic compounds

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